

ANKIT KUMAR TIWARI

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SUMMARY

Dedicated Computer Science Engineering student at VIT Bhopal with a strong focus on Machine Learning and Deep Learning. Proficient in developing AI-driven solutions, including real-time computer vision systems and containerized ML pipelines. Active competitive programmer on Codeforces and LeetCode, passionate about solving complex algorithmic challenges and building scalable, data-centric applications.

TECHNICAL SKILLS

- **Data Science & ML:** Python, Pandas, Numpy, scikit-learn, Tensorflow, Pytorch, Feature Engineering, Model Evaluation, Clustering, Statistical Analysis, Regression
- **AI & NLP :** LangChain, RAG Pipelines, Transformers, LLM APIs(OpenAI/ Anthropic/Google), Text Classification, NLP preprocessing
- **Data Visualisation:** Matplotlib, Seaborn, Plotly, Leaflet.js, REST API Dashboards, Interactive Charts and Reports
- **Automation & Dev:** FastAPI, React.js, Next.js 14, Node.js, ReactNative, MySQL, Flask, Rest APIs, JSON, Git/Github

EDUCATION

VIT Bhopal University

Aug 2024-May 2028

B.Tech in Computer Science & Engineering — CGPA: 8.78 / 10.0

Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, DBMS, Operating Systems, Computer Networks, Statistics, Object-Oriented Programming

PROJECTS

AI-Powered Government Tender Evaluation System | Python, Gemini API, Multi-Agent AI, OCR, FastAPI, React

- Built a multi-agent AI system where four agents work in a pipeline — one extracts eligibility requirements from tender PDFs using OCR, another parses bidder documents, a third evaluates bids against criteria with clear pass/fail verdicts, and a fourth flags uncertain results for human review.
- Integrated Google Gemini 2.0 Flash API to analyse scanned government documents, return structured JSON responses, and handle Indian currency formats (Crore, Lakh); added confidence scoring so low-quality OCR results get automatically flagged instead of silently passing.
- Developed the full-stack application with a FastAPI backend (REST API, PDF report generation) and a React dashboard where users can upload documents, view evaluation results, review flagged items, and download audit reports — all from a single interface.
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YouTube Video Chatbot | Python, LangChain, FAISS, RAG, HuggingFace Llama 3, Streamlit

- Built a RAG pipeline that extracts YouTube transcripts, generates semantic embeddings (BGE-large), and stores them in a FAISS vector store for similarity-based context retrieval.
- Integrated Meta Llama 3 via HuggingFace API with LangChain prompt templates and parallel runnables to deliver context-grounded conversational Q&A over video content.
- Deployed an interactive Streamlit web app with modular architecture and session-state management, enabling end-to-end video-to-answer functionality from a single URL input.

AI-Powered Career Coaching System | Python, HuggingFace, Gemini API, Docker, Next.js, RAG

- Fine-tuned a DistilBERT NLP model with PyTorch to categorize user text into 65 distinct career paths.
- Engineered a RAG-based backend using the Gemini API to provide personalized career advice and generate interview questions based on user input.
- Built the frontend chatbot interface using Next.js and containerized the full-stack application with Docker Compose.

EXTRACURRICULARS & PARTICIPATIONS

- **National Hackathon Finalist** — SIH & India Innovates
- **Core Member**, Linpack Club, VIT Bhopal
- **Interests:** Cricket and Technical Problem Solving.
- **Competitive Programming:** Codeforces Rating(1000+)